

# MU-HUA (MAURICE) WANG

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## RESEARCH INTEREST

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**Physical assistive robotics for long-term care.** I want to build robots that perform contact-rich care tasks on a bedridden body, starting with diaper changing. The task demands three things at once: perceiving a body that will not cooperate, knowing enough about that person to move them without harm, and handling deformable material safely against skin. My work covers the first two and has begun the third: a clinically validated markerless motion capture system, studies of what a robot must know about a body, and a diaper-changing simulation platform in progress.

## EDUCATION

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**National Yang Ming Chiao Tung University** — M.S., Biomedical Engineering

Entered Feb 2025 · on approved leave of absence since May 2025, enrollment retained · researching full time

**National Yang Ming Chiao Tung University** — B.S., Biomedical Engineering · Sep 2020 – Jun 2024

– GPA 3.72 / 4.30 · final two years 4.07 / 4.30 · IELTS 8.0 (R 9.0, L 8.5, S 7.5, W 7.0)

## RESEARCH EXPERIENCE

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Project pages, videos and full manuscripts for everything below: [maurice1128.github.io](https://maurice1128.github.io)

### Care-Robot Simulation Platform — independent · 2026 – present · in progress

**Measuring Whether a Care Robot Hurts the Patient:** robot-assisted diaper changing and perineal cleaning

- A MuJoCo simulation that scores every run on three axes at once: is the garment on, is the perineum cleaned, is the patient injured. The patient is a physically settled SMPL-X body with pelvic bones under the skin and a soft-tissue model whose solver matches closed-form tests to  $1.8 \times 10^{-5}$ .
- Measured so far, in simulation only: a brief closed around the settled body with all four tabs fastened; 0.93 of the main perineal region wiped at least three times under a criterion fixed in advance; after 285 min supine, 30 kPa foam against a rigid bed lowers skin pressure by about 15% but cuts tissue above the strain level from 69 to 4 cm<sup>3</sup>. Video on the website.

### Robot Learning and Embodied AI — independent · Oct 2025 – present

Sole author on five manuscripts, **four submitted**. Each result is tested against controls built to break it, and reports what it did not show.

**Gentle by Knowing: How Much Must a Care Robot Know About a Person to Lift Their Arm Safely?** — submitted, *Machine Learning (Springer)*

- Musculoskeletal simulation of a robot lifting the forearm of simulated care recipients with restricted shoulders and elbows; controllers differ only in what they know about the person. **No force-threshold guard cut harmful outcomes by significantly more than 5 percentage points:** 95–100% of harmful outcomes began in the first attempt. A policy learned offline in simulation cut them from 110 to 25 of 300 simulated persons at a similar number of safe completions (174 vs 177), from a care record with 20° error.

**What a Motor Prior Is Worth: A Body Model and a Teacher Compared to Convergence and Through Withdrawal** — submitted, *IEEE TCDS*

- A babble-learned frozen body model against a distilled teacher, one SAC learner, six muscle-driven bodies, trained to a checked plateau. After convergence a learner that lost its teacher is never detectably worse than one never guided; the body model beats model-free learning on four of six.

**Modularity Makes Dexterous Hands Learn Faster, Not Better** — submitted, *IEEE RA-L*

- Tested both promises of a modular motion model on a simulated Shadow Hand, main comparisons fixed before the runs. A keyframe interface learns held-out skill sequences far better early (0.59 vs 0.22 success at 1.5 M frames), a continuous latent catches up (0.49 vs 0.44 at 5 M), and separately trained skills do no better than one flat policy (45 vs 61 of 400 episodes).

**A Hand-Motion Dataset Can Test Compositional Generalisation Only When Its Labels Describe the Scored Motion** — submitted, *TMLR*

- Calibrated the held-out-combination penalty on four public hand datasets against zero-truth, planted and permuted-label controls. On OakInk-Image, when correct labels described only half of each trajectory the penalty

fell from +17.0% to +3.9%, not distinguishable from the control; a penalty predicted on TACO before its data were opened was found (+10.6%).

**In draft:** *Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer*, where an algebraic identity rules out a family of proposals before any experiment runs.

## RGB Multi-View 3D Markerless Motion Capture — NYCU · Oct 2023 – Sep 2025

Research Assistant, then Team Lead · Advisor: Prof. Hui-Ting Shih · the perception layer a robot over a bed needs

- **Built the real-time pipeline end to end:** GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT with occlusion handling; and **ROWV**, an outlier-rejection algorithm beating KNN and RANSAC on this problem. **Validated** against VICON over 21 participants, plus 104 walking trials on a nine-camera ring.
- **Asked how much sensing error the downstream decision can absorb:** in forward-dynamic simulation, identification of a weak muscle group falls  $0.845 \rightarrow 0.549$  as joint-angle error grows  $0^\circ \rightarrow 12^\circ$ .
- **Ran the collaborations and the field work:** 2 hospital MOUs (Taipei Veterans General; Taipei City Hospital), 200+ stroke and knee-pain patients in Vietnam, 40+ at Kaohsiung Chang Gung, 20+ elderly at a daycare centre, amputee trials with NYCU Physical Therapy.
- **Shipped the product** and led a 5-person team: PyQt6 clinician GUI, SQL and cloud pipeline with ICD-10 coding, Unreal Engine rehabilitation game; 6 media features.

## Behavioural Trajectory Modelling — Taoyuan General Hospital · Jan 2023 – present

Advisor: Dr. Tso-Jeng Wang. Mapped methadone dose trajectories to a polar angle, then clustered with a bivariate von Mises mixture on the torus. 9,900 urine tests from 552 patients.

## EEG and EMG Sleep–Epilepsy Analysis — NYCU · Oct 2022 – Oct 2023

Advisor: Prof. You-Yin Chen. Built a pipeline processing 24-hour recordings in 3 minutes with automatic WAKE / NREM / REM staging; designed a surgical modification that raised EMG signal-to-noise.

## PUBLICATIONS

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### Peer-reviewed conference papers

1. **Wang, M.-H.**, et al. *Markerless Gait Analysis for Lower Back Pain*. SEMBA 2025. **(first author)**
2. Huang, L.-W., **Wang, M.-H.**, et al. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users*. ASB 2025.
3. Huang, A.-Y., **Wang, M.-H.**, et al. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle*. SEMBA 2025.
4. Chang, C.-W., **Wang, M.-H.**, et al. *Central Thalamic Deep Brain Stimulation and Sleep in an Alzheimer's Model*. ISMRM 2024.
5. Chen, B.-H., et al. incl. **Wang, M.-H.** *Robust Self-Occlusion Mitigation in Multi-View 3D Markerless Motion Capture*. ICMST 2024.

**Submitted** (sole author, 2026). The four above: *Machine Learning* (Springer) · IEEE TCDS · IEEE RA-L · TMLR.

**Drafts** (on the website). One in robot learning, three in clinical biomechanics, one in behavioural data science.

## TEACHING

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**Physical Science, Grade 8** — Tamkang High School · Feb – Aug 2026

Class 8 rose to 2nd of eight classes, its margin over the school average widening  $+5.1 \rightarrow +11.8$  points. **The gain came from the bottom:** its bottom-12% cutoff finished 28 points above the school's, up from 14.

## SKILLS AND AWARDS

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**ML / Robotics** — Python, MATLAB, C/C++ · PyTorch · SAC, PPO, model-based RL, policy distillation, offline policy learning · MuJoCo, MyoSuite, Shadow Hand, SMPL-X, inverse kinematics

**Systems / Statistics** — CuPy, Numba, PyQt6, SQL · paired designs, permuted-label controls

**Awards** — 2nd Place, Youth AI Startup Hackathon, Ministry of Digital Affairs (2023) · 1st Place, Elderly Health & Technology, NCNU (2023)